



Article on Walmart Database.

Yash Parshotam Panjwani | 10/17/2024

Table of content

1. Introduction.
 - 1.1 Overview of the Open-Source Data.
 - 1.2 Purpose and Goals
2. Research Question
 - 2.1 Key Question 1
 - 2.2 Key Question 2
 - 2.3 Key Question 3
3. Metrics and KPI's
 - 3.1 Total Sale
 - 3.2 Total Profit
 - 3.3 Item Sold
4. Strategic objective.
5. Data Model.
 - 5.1 Overview of data.
 - 5.2 Entity-Relationship Diagram (ERD)
6. Results.
7. Conclusion.
8. Appendices.

1. Introduction

1.1 Overview of the Open-Source Data

This article examines sales data sourced from Data World, focusing on Walmart's performance across various locations in the United States. The dataset includes key sales metrics and product details, offering insights that can help drive profitability. Notably, this data covers the period from January 1 to January 10, 2023. By analyzing this data, Walmart aims to identify trends and opportunities that enhance its profit margins and optimize its overall business strategy.

1.2 Purpose and Goals

Walmart plans to increase its profit margins by focusing on products that bring in higher profits and offering special deals to attract higher-income customers. At the same time, they'll keep serving their regular value-focused shoppers. This way, Walmart can appeal to more people, boosting profits from higher-end products while still keeping prices low for their core customers.

2. Research Question.

- 2.1 How can we raise profit margin without losing sales?
- 2.2 Which customer segments drive the most profit?
- 2.3 Are discounts helping or hurting profits?

3. Metrics and KPI's

3.1 Total Sale

Total sales represent the overall revenue Walmart generates from selling products over a specific period. It gives a clear picture of how much money is coming in from all customer purchases. This number is important because it shows the scale of Walmart's business and its ability to attract and retain customers

3.2 Total Profit

Total profit refers to how much money Walmart keeps after deducting all the costs involved in running the business, like buying products, paying employees, and maintaining stores. It's a key measure of financial health, showing how well Walmart is converting sales into earnings.

3.3 Total Item Sold

The "Items Sold" metric counts the total number of products that have been purchased by customers. This KPI helps Walmart understand how many individual units are moving off the shelves, which is useful for tracking product demand and ensuring the right stock levels.

4. Strategic Objectives

Walmart's goal is to grow its business by increasing sales, making more profit, and ensuring the right products are always available. By focusing on selling high-profit items, running its operations more efficiently, and reaching a wider range of customers, Walmart aims to boost profits while still offering value to its loyal shoppers. This strategy helps Walmart stay competitive and meet customer demand.

5. Data Model

5.1 Overview of data model.

1. **Geography Table:** Contains geographical data for regional analysis.
2. **Customer Table Lookup:** Holds customer information for segmentation and behavior analysis.
3. **Order Table Lookup:** Captures order details for sales trend examination.
4. **Product Table Lookup:** Lists product information to analyze sales performance.
5. **Fact Table:** Consolidates transaction-level data for comprehensive sales analysis.

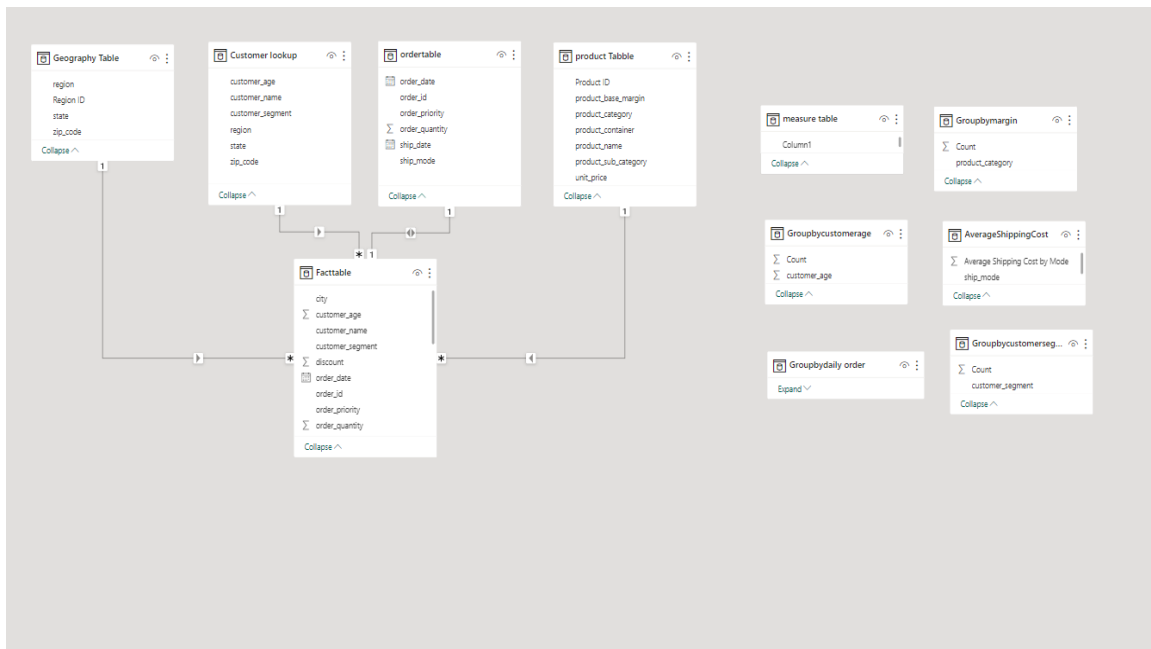
Measures

- **Total Profit:** Overall profitability.
- **Total Sales:** Total revenue generated.
- **Products Sold:** Quantity of products sold daily and in total.

Group By Insights

- **Customer Age:** Reveals purchasing patterns across age groups.
- **Daily Order:** Identifies peak sales periods
- **Margin:** Shows profitability of different categories.
- **Average Shipping Cost:** Analyzes average shipping costs.
- **Customer Segment:** Allows for targeted marketing strategies based on customer categories.

5.2 Entity Relationship Diagram.



6 Results.

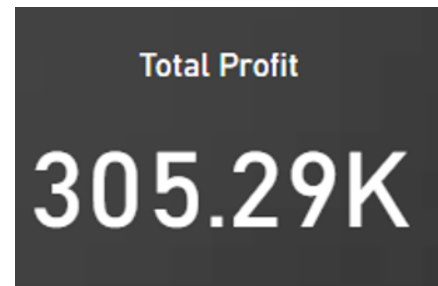
6.1 Total Sale.

The total sales amount shown is **1.61 million**, representing the total sales made by Walmart over a 9-day period across different states



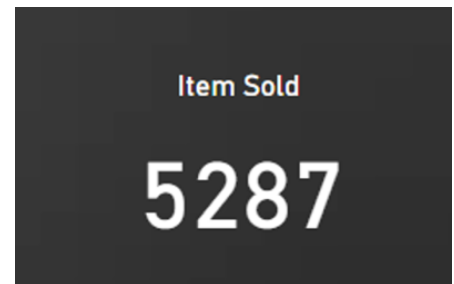
6.2 Total Profit.

The total profit shown is **305.29 thousand**, which represents the overall profit generated by Walmart from sales over a 9-day period across different states



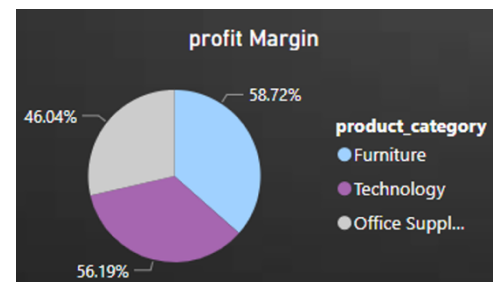
6.3 Item Sold.

The total number of items sold is **5,287**, representing the quantity of products purchased by customers across different states from Walmart from sales over the 9-day period.



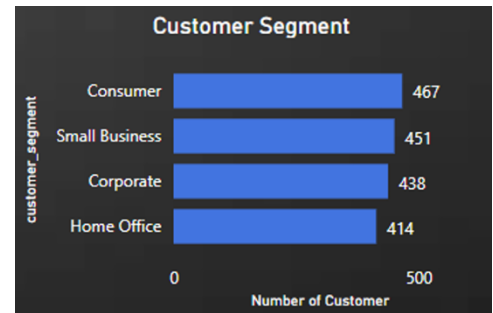
6.4 Profit Margin.

The pie chart I created shows the profit margins for three product categories. Furniture has the highest margin at **58.72%**, followed by technology at **56.19%**, and office supplies with a margin of **46.04%**.



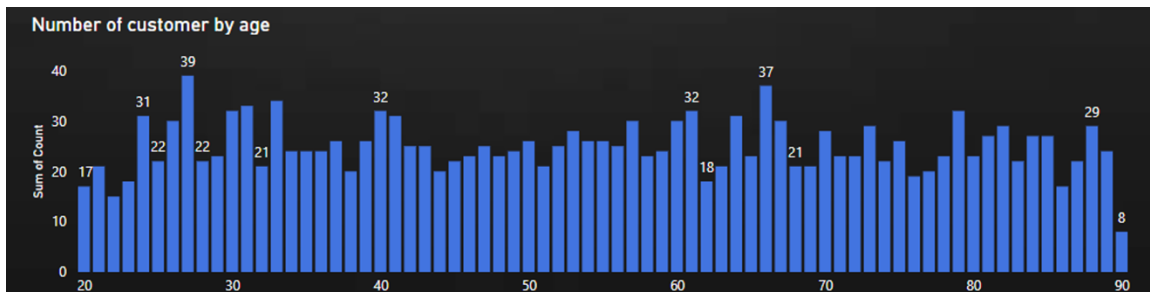
6.5 Customer Segment.

The bar graph I created displays the number of customers in different segments. The **consumer** segment has the highest number with **467** customers, followed by **small business** with **451**, **corporate** with **438**, and **home office** with **414** customers.



6.4 Number of customers by age.

The bar graph displays the ages of customers who placed orders, highlighting our customer demographic. The most orders came from the age group of 30 to 39 years, indicating a strong engagement from this segment.



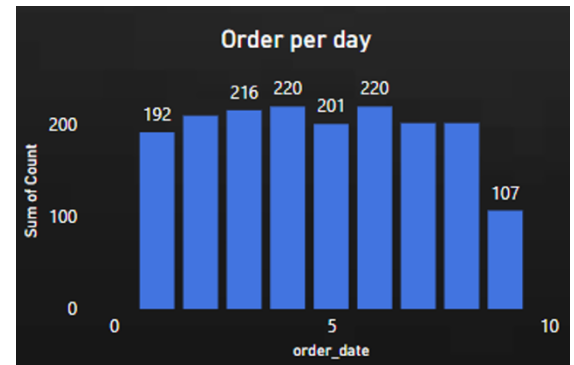
6.6 Shipping cost.

The bar graph shows the average costs for three shipping methods. Express Air costs an average of \$1.2K, making it the most expensive option. Both Regular Air and Delivery Truck services cost about \$0.9K on average, making them more affordable choices.



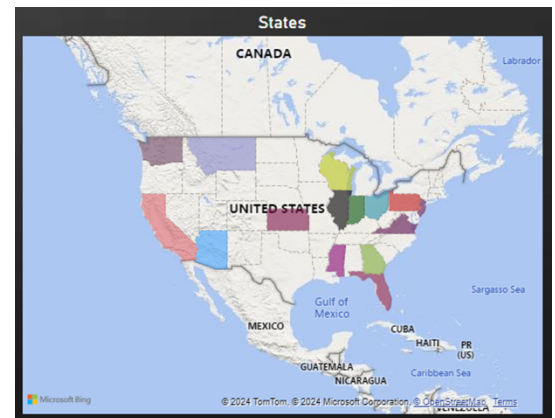
6.7 Order per day

The bar graph shows the number of orders we received each day. This visual highlights trends in customer activity and order volume.



6.8 States

The map shows the provinces where our orders came from, giving us a clear look at where our customers are located. By seeing this information, we can understand which areas have the most demand for our products, helping us improve our marketing and delivery strategies.



6.9 Day slicer.

I created a day slicer that lets you pick specific dates. When you select a date, the sales, profit, and order graphs change to show the relevant data. It makes it simple to see trends for that time.

Day
☐ 1
☐ 2
☐ 3
☐ 4
☐ 5
☐ 6
☐ 7
☐ 8
☐ 9

7. Conclusion.

In conclusion, by examining key performance indicators like total sales, total profit, items sold, and daily orders, we gain valuable insights into how various customer segments are performing. This analysis enables Walmart to identify which segments are contributing most to revenue and highlights areas where improvements can be made. Understanding these dynamics allows the company to optimize its sales strategies and enhance profitability, ultimately leading to better customer satisfaction and business growth.

8. Appendix.

Dashboard 1.



Dashboard 2.



Dashboard 3.

